Exercise 11.1

Consider a system

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx\end{aligned}$$

where $x \in \mathbb{R}^n$ is the state, $u \in \mathbb{R}$ is the input, and $y \in \mathbb{R}$ is the output. The system is controlled by a PID controller

$$u(t) = -K_p \left(y(t) + T_d \dot{y} + \frac{1}{T_i} \int_0^t y(\tau) d\tau \right)$$

- Write an expression for the controller dynamics on state space form.
- Write an expression for the closed-loop dynamics on state space form.

Exercise 11.2

Consider the system

$$\begin{aligned}\dot{x} &= \begin{bmatrix} 1 & 2 \\ 0 & 4 \end{bmatrix} x + \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} u + w \\ y &= \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x + v\end{aligned}$$

where $x \in \mathbb{R}^2$ is the state, $u, w \in \mathbb{R}^2$ is the input, and $y, v \in \mathbb{R}^2$ is the output. In addition

$$\mathcal{E}\{ww^T\} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad \mathcal{E}\{vv^T\} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

- Compute an optimal estimator (gain L) for the system (use `lqe`).
- Simulate the system and estimator in Simulink, and plot the state estimation error.